

rv32im CPU chip

clk	pc	instr	a0
t	x	ecall	...
...	...	$t_0=0x135$	...

byte range chip

byte
0..255
...

Griffin controller chip

(1 row / syscall)

clk	state_addr	memory[16]	final_value[16]
t	P	in-state	out-state

cross-table lookup
InteractionKind=22
(initial/final state)

Griffin per-round chip

(14 rows / syscall = NB_ROUNDS)

is_real	round_idx	state_before[4]	sbox/MDS/rc_add	rc_add.result
1	0	in	...	→ row 1
1	1	(= prev rc_add)	...	→ row 2
⋮	⋮	⋮	⋮	⋮
1	13	...	...	out

next.state_before = rc_add.result

Legend (after ePrint 2023/1032, 2025/2204): - → lookup relation ↔ permutation relation.

Columns are the implemented GriffinFp192Chip/ControlCols of this thesis; the precompile is one SPI chip-component bound to the CPU trace by the lookup argument.